

Wazuh SIEM Deployment Session

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Session Summary

Task	Status
Wazuh LXC (CT104) created on Servers VLAN	Done
Wazuh all-in-one installed (v4.11.2)	Done
Dashboard accessible at https://192.168.10.71	Done
Wazuh agent installed on Vaultwarden (CT102)	Done
Version mismatch diagnosed (agent 4.14.6 vs manager 4.11.2)	Done
Upgrade attempted — cert/password issues encountered	Done
Snapshot rollback executed successfully	Done
Passwords reset via wazuh-passwords-tool	Done
Dashboard restored and accessible	Done
Agent registration — pending next session	Pending
OPNsense syslog configuration	Pending
Manager upgrade to 4.14.x (official docs method)	Pending

1. Wazuh Architecture

Wazuh has three components deployed as an all-in-one install on CT104:

Component	Purpose	Port
Wazuh Manager	Receives and analyzes logs from agents	1514 (agents), 1515 (registration)
Wazuh Indexer	Stores log data (OpenSearch/Java)	9200 (localhost only)
Wazuh Dashboard	Web UI for viewing alerts and events	443

Agent vs Syslog: Agents are installed on Linux machines for deep visibility (process monitoring, file integrity, active response). Syslog is used for devices that can't run an agent — like OPNsense (FreeBSD).

2. LXC Creation (CT104)

2.1 Specs

Setting	Value
CT ID	104
Hostname	wazuh
IP	192.168.10.71 (Servers VLAN, tag 10)

RAM	8GB
CPU	4 cores
Disk	50GB
Template	Debian 12
Privileged	Yes (required for Wazuh indexer)

Creation command:

```
pct create 104 local:vztmpl/debian-12-standard_12.12-1_amd64.tar.zst \ --hostname wazuh
--storage local-lvm --rootfs local-lvm:50 \ --memory 8192 --cores 4 \ --net0
name=eth0,bridge=vbr0,ip=192.168.10.71/24,gw=192.168.10.1,tag=10 \ --unprivileged 0 --features
nesting=1 --start 1
```

Why Servers VLAN: Wazuh is a workload/service, not management infrastructure. Placing it on Servers VLAN limits blast radius — if compromised, attacker is isolated from Proxmox and OPNsense on Management VLAN.

Why privileged: Wazuh indexer (OpenSearch/Java) requires low-level system access that unprivileged containers block.

3. Wazuh Installation

3.1 Download and Configure

```
curl -sO https://packages.wazuh.com/4.11/wazuh-install.sh
curl -sO https://packages.wazuh.com/4.11/config.yml
```

Edit config.yml — set all three node IPs to 192.168.10.71:

```
nodes: indexer: - name: node-1 ip: "192.168.10.71" server: - name: wazuh-1 ip: "192.168.10.71"
dashboard: - name: dashboard ip: "192.168.10.71"
```

3.2 Run Installer

```
bash wazuh-install.sh -a
```

The -a flag = all-in-one. Install takes 15-20 minutes. Auto-generates credentials at the end — save them to Vaultwarden immediately.

3.3 Verify Services

```
systemctl status wazuh-manager wazuh-indexer wazuh-dashboard
ss -tlnp | grep -E "443|9200|1514|1515"
```

Port	Process	Purpose
443	node (dashboard)	Web UI
1514	wazuh-remoted	Agent log ingestion
1515	wazuh-authd	Agent registration/auth
9200	java (indexer)	OpenSearch API — localhost only

Security note: Port 9200 bound to 127.0.0.1 only — correct by design. OpenSearch API should never be exposed to the network.

4. Wazuh Agent — Vaultwarden (CT102)

4.1 Install

Install via official Wazuh repo with GPG verification:

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | gpg \ --no-default-keyring \ --keyring  
gnupg-ring:/usr/share/keyrings/wazuh.gpg --import chmod 644 /usr/share/keyrings/wazuh.gpg echo  
"deb [signed-by=/usr/share/keyrings/wazuh.gpg] \ https://packages.wazuh.com/4.x/apt/ stable  
main" \ | tee /etc/apt/sources.list.d/wazuh.list apt update && apt install wazuh-agent -y
```

4.2 Configure Manager IP

Edit `/var/ossec/etc/ossec.conf`:

```
<client> <server> <address>192.168.10.71</address> <port>1514</port> <protocol>tcp</protocol>  
</server> </client>  
  
systemctl enable wazuh-agent && systemctl start wazuh-agent
```

4.3 Version Mismatch Issue

Agent installed as v4.14.6 but manager is v4.11.2. Wazuh requires agent version $\leq$ manager version. Agent log showed:

```
ERROR: Agent version must be lower or equal to manager version (from manager)
```

Fix: Upgrade the manager (not downgrade the agent). Manager must always be equal or ahead of agents. Attempted upgrade encountered cert/password issues — rolled back via snapshot. Proper upgrade pending next session using official Wazuh upgrade docs.

5. Troubleshooting Log

Issue: sudo not installed

Fix: apt install sudo -y — script written for universal Linux compatibility, uses sudo even when running as root

Issue: Dashboard cert missing after upgrade

Fix: EACCES/ENOENT on dashboard-key.pem — upgrade wiped certs directory. Regenerated with `wazuh-install.sh -g`

Issue: Cert ownership wrong

Fix: wazuh-dashboard process couldn't read certs — `chown -R wazuh-dashboard:wazuh-dashboard /etc/wazuh-dashboard/certs/`

Issue: Indexer password mismatch

Fix: New certs generated new passwords but indexer still had old ones — unauthorized on curl to localhost:9200

Issue: Original password lost

Fix: Saved new password over old one in Vaultwarden before confirming rollback worked

Issue: Snapshot rollback

Fix: `pct stop 104 && pct rollback 104 pre-wazuh-upgrade && pct start 104` — restored clean state

Issue: Password reset

Fix: `/usr/share/wazuh-indexer/plugins/opensearch-security/tools/wazuh-passwords-tool.sh --change-all`

6. Key Lessons

- Save all credentials to Vaultwarden immediately — never overwrite without confirming the old one still works
- Snapshot before every major change — pct snapshot takes seconds and saves hours
- Read logs before touching anything — journalctl, systemctl status, and ss -tlnp diagnose most issues

- Upgrade manager not agents — manager must always be $\geq$ agent version
 - File permissions matter — EACCES means wrong ownership, not wrong config. Check chown and chmod first
 - Package signing vs curl|bash — GPG-verified repos are safer than piping scripts directly to bash
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7. Next Session

- Upgrade Wazuh manager to 4.14.x using official upgrade guide
 - Register Vaultwarden agent with manager
 - Install agents on CT103 (Homepage) and pve1 (Proxmox host)
 - Configure OPNsense syslog to forward firewall logs to Wazuh
 - Add wazuh.mystikal.dev subdomain via Caddy
 - Review first alerts in the dashboard
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